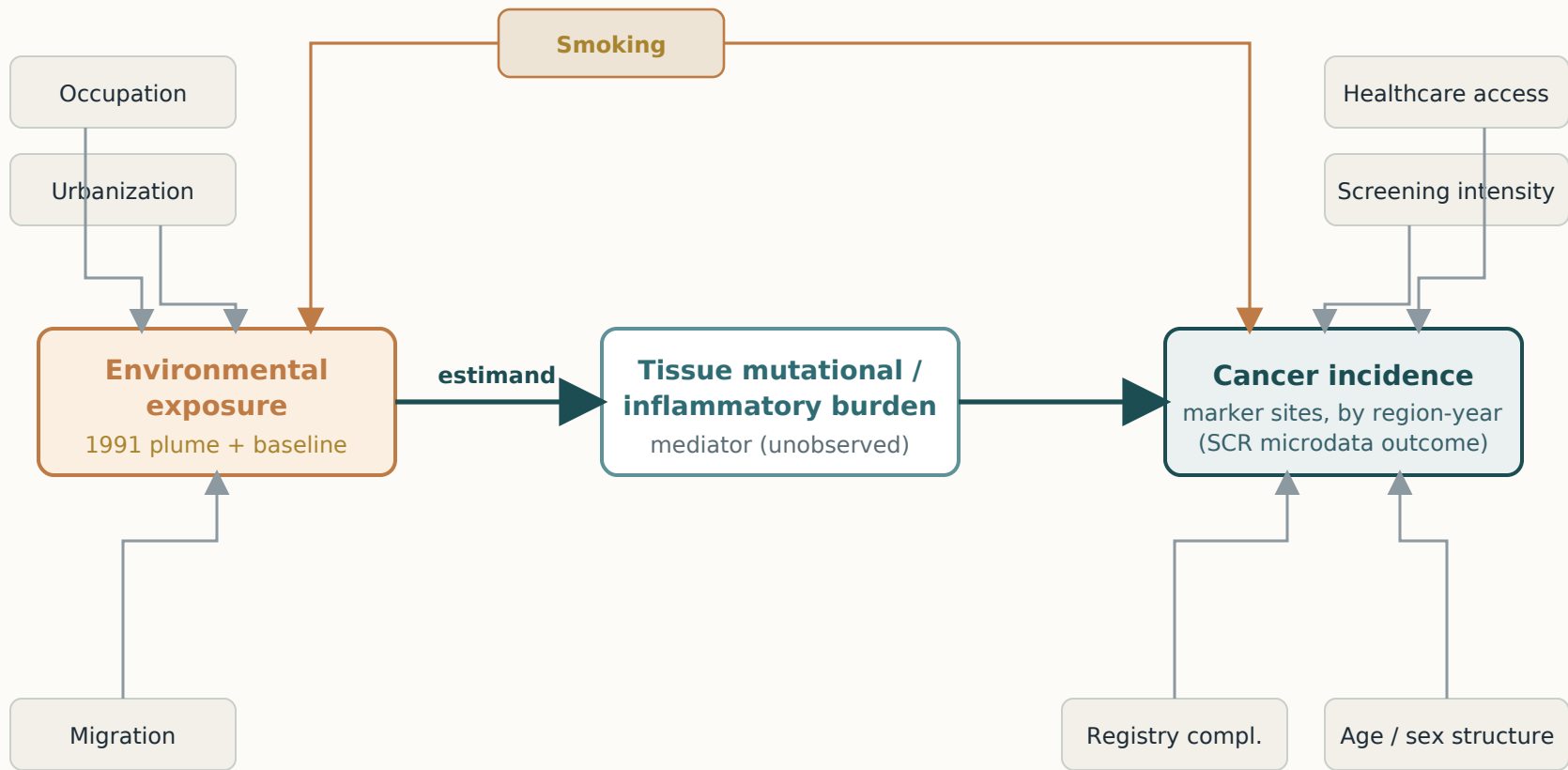




Teal: causal path to be estimated. Grey: confounders / ascertainment factors, each a separate edge, entered as covariates, fixed effects, or negative-control comparisons. Smoking (clay) is the key dual-path confounder; a never-smoker lung-cancer subcohort blocks it directly.